

IDEATION PHASE

Empathize & Discover

Date: 19 June 2026

Team ID: -----

Project Name: AI-Enhanced Intrusion Detection System

Maximum Marks: 4 Marks

Empathy Map Canvas

An empathy map helps understand the needs, challenges, goals, and behavior of the target users. For the AI-Enhanced Intrusion Detection System, the primary users are:

- Security Analysts
- Network Administrators
- IT Managers
- Cybersecurity Teams

Understanding their concerns helps build a more effective cybersecurity solution.

User Persona

Primary User

Name: Security Analyst

Age: 25-45 Years

Role: Monitor network traffic and investigate security threats

Goals:

- Detect cyber attacks quickly
 - Reduce false alarms
 - Improve network security
 - Automate threat monitoring
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Empathy Map

Diagram 1: Empathy Map Canvas

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WHAT DOES THE USER  
THINK & FEEL?

- Concerned about cyber threats
- Wants faster threat detection
- Needs accurate alerts
- Wants secure network operations



WHAT DOES USER HEAR? WHAT DOES USER SEE?

- Security team warnings • Network traffic logs • Industry cyber alerts • Attack notifications • Compliance requirements • Dashboard statistics • Management expectations • Security incidents

WHAT DOES USER SAY & DO?

- Monitor network traffic • Investigate suspicious activity • Generate security reports • Configure security tools

PAIN GAIN

- Too many false alerts • Accurate detection • Manual monitoring • Automated alerts • Slow incident response • Faster response time • Unknown threats • Better security visibility

**\*\*Figure 1: Empathy Map for Security Analyst\*\***

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# User Perspective Analysis

## What Does the User Think & Feel?

- Network security is critical.
- Cyber attacks are becoming more sophisticated.
- Manual monitoring is time-consuming.
- Accurate detection is essential for business continuity.

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### ## What Does the User Hear?

- Security compliance requirements.
- Alerts from threat intelligence feeds.
- Recommendations from cybersecurity experts.
- Organizational security policies.

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### ## What Does the User See?

- Network traffic logs.
- Security dashboards.
- Intrusion alerts.
- Threat reports and statistics.

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### ## What Does the User Say & Do?

- Reviews suspicious network activity.
- Responds to security incidents.
- Generates security reports.
- Monitors intrusion detection dashboards.

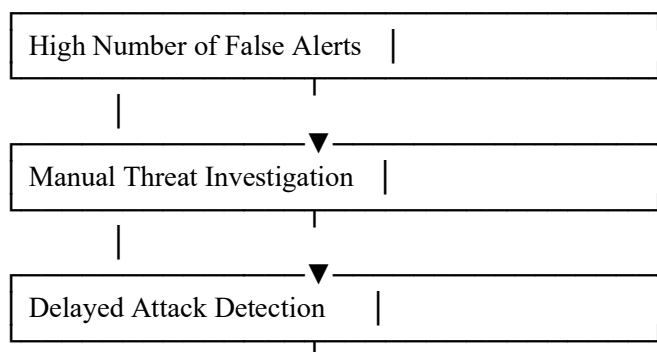
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## # Pain Points & Challenges

### ## Diagram 2: Pain Point Analysis

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#### PAIN POINTS



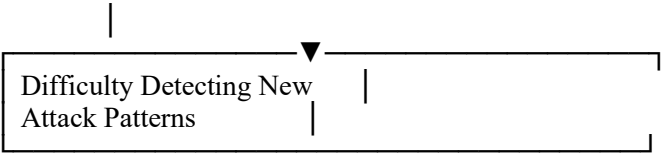


Figure 2: User Pain Points

## Expected Gains

Diagram 3: Gain Analysis

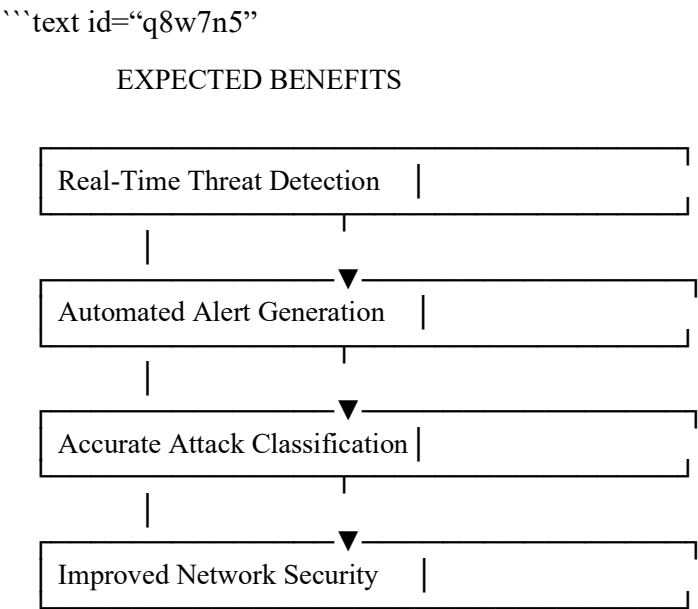


Figure 3: Expected User Gains

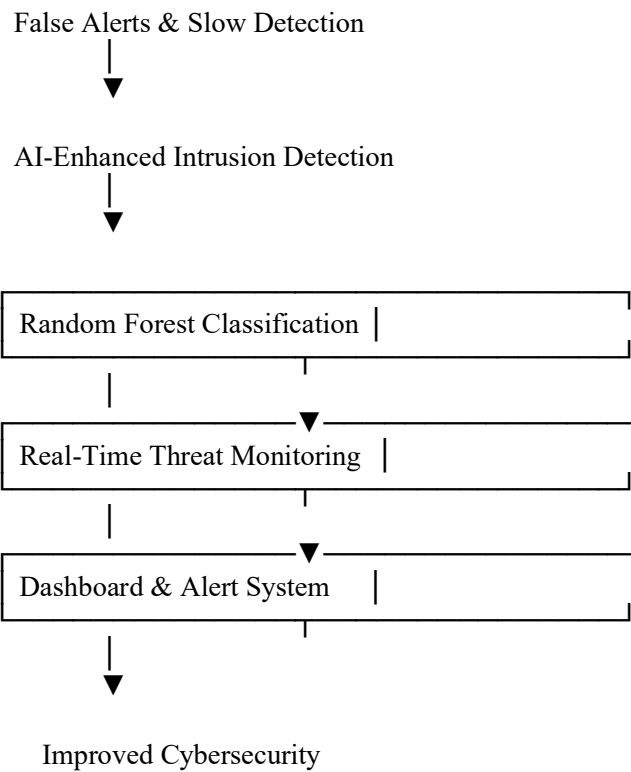
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# Solution Mapping

Diagram 4: Empathy to Solution Mapping

USER PROBLEMS





**Figure 4: User Needs to Solution Mapping**

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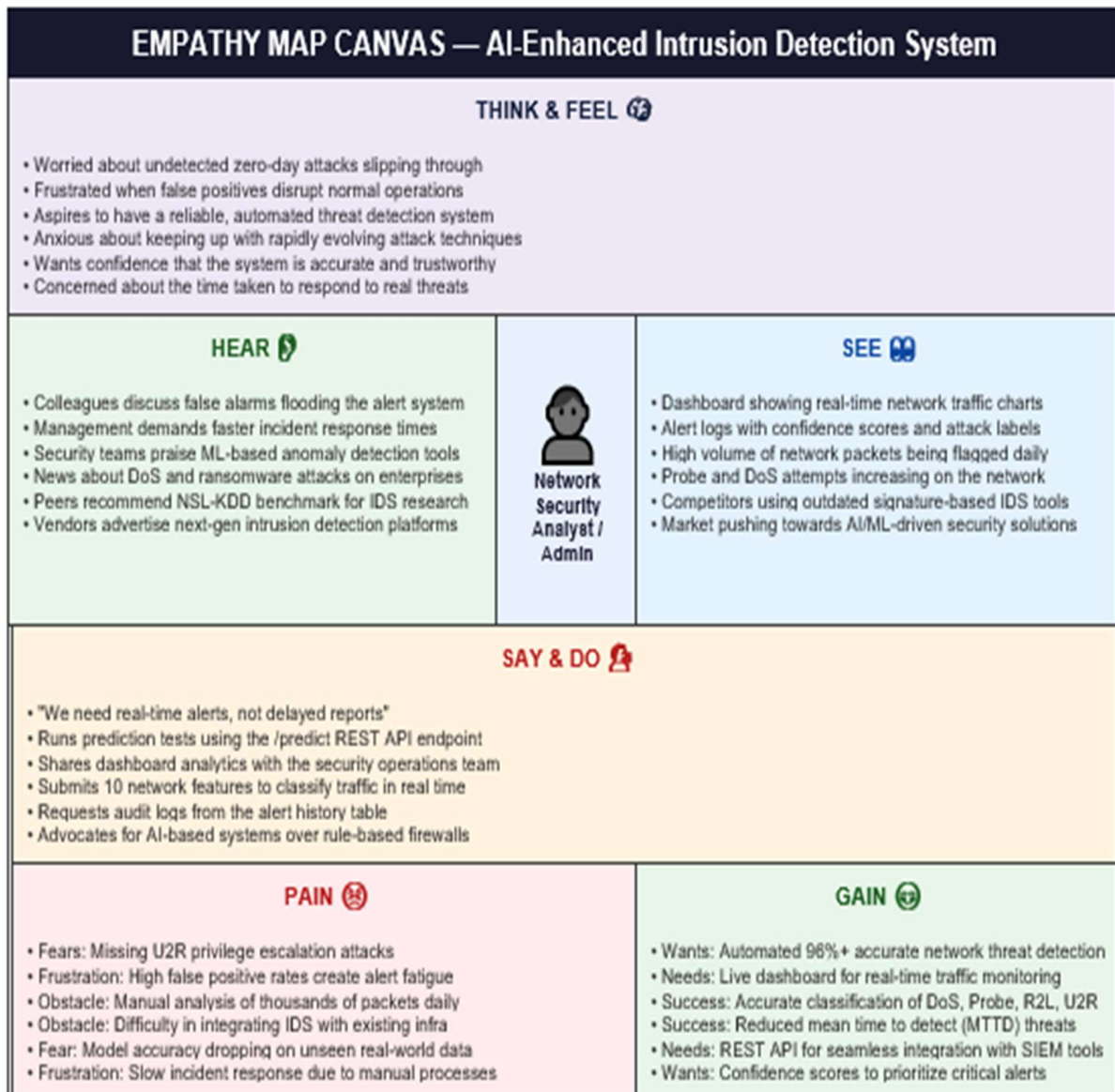
## Key Insights

### User Needs

- Faster intrusion detection
- Reduced false positives
- Real-time monitoring
- Easy-to-use dashboards
- Automated threat analysis

### Solution Features

- AI-Based Intrusion Detection
  - Random Forest Machine Learning Model
  - Automated Alert Logging
  - Dashboard Visualization
  - REST API Support
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## Conclusion

The Empathy Map Canvas helped identify the needs, frustrations, and expectations of security analysts and network administrators. The insights gathered guided the development of the AI-Enhanced Intrusion Detection System, ensuring that the final solution effectively addresses real-world cybersecurity challenges through automation, accuracy, and real-time threat detection.